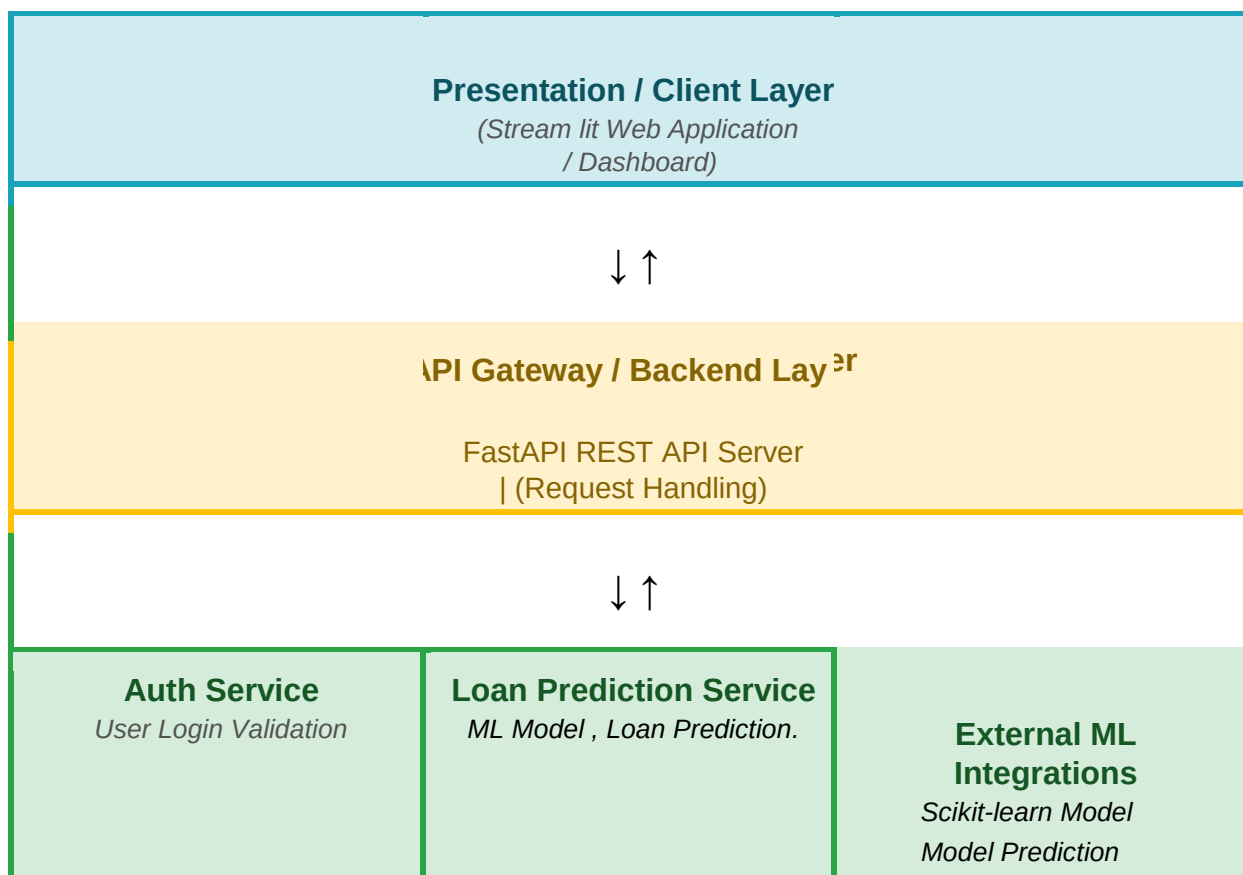


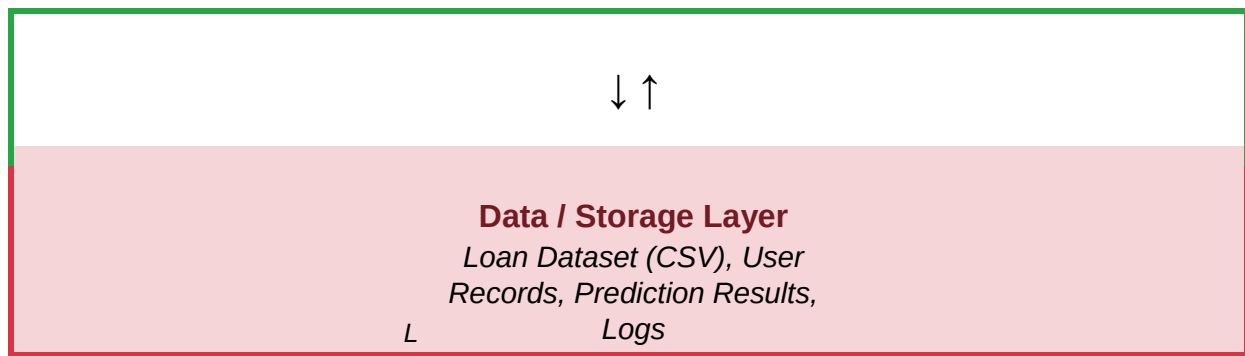
Solution Architecture

Date	28 June 2026
Team ID	SWTID-2026-9068
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.





Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a simple web interface for entering applicant details and displaying loan approval results.	Streamlit, HTML, CSS
API Gateway	Receives requests from users, validates input, and forwards them to the ML prediction service.	FastAPI, Uvicorn
Core Logic/Prediction Service	Processes applicant data, performs preprocessing, and predicts loan approval using the trained ML model.	Python, Scikit-learn, Pandas, NumPy
Database / Storage	Stores loan dataset, prediction history, user inputs, and application logs.	CSV Files, SQLite (or CSV Storage)
Machine Learning Model	Predicts whether the applicant is eligible for a loan based on financial information.	Scikit-learn, Joblib
Testing Layer	Tests APIs, prediction accuracy, and application functionality.	PyTest

